

LSE (LemonSeed-Engine) + FlashPrefill V2

Long-context speed and program correctness

Baseline · MTP3 · DFlash2
Off versus alpha 0.1

1,368 completed tests · Standard, 16K and 32K

Results at a glance

51–55% faster prompt processing at 32K

Baseline +54.7% · MTP3 +51.2% · DFlash2 +54.7%

Matching pass outcomes at Standard and 32K

At 16K: Off passed 32/32; alpha 0.1 passed 31/32.

HumanEval/97 changed from pass to fail in each inference mode.

Decode throughput stayed close; decode attention remains dense.

Pass rate is functional program correctness; it does not measure all forms of model quality.

What was measured

164 standard tasks + 32 at each long context

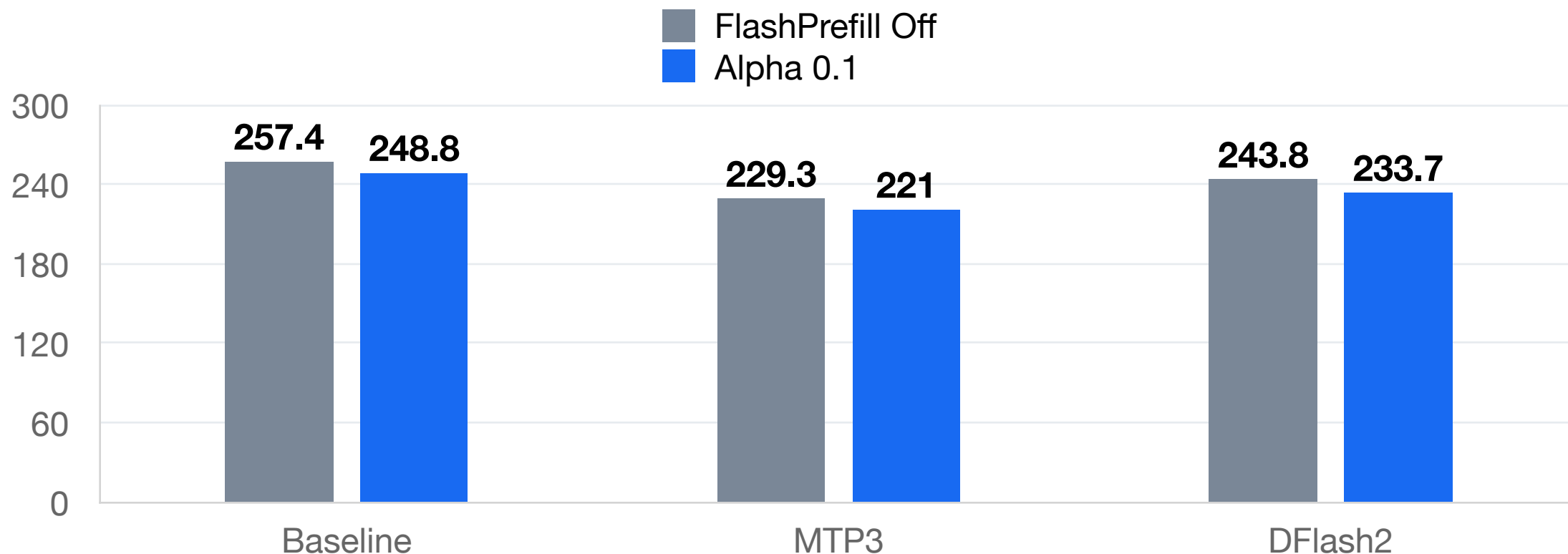
Each of six configurations ran the same prompts: three inference modes, each with FlashPrefillV2 Off and alpha 0.1.

One greedy completion; every test must pass

Official base and extended checks. Output limit: 2,048 tokens.
Prefill batch: 1,024. BF16 K/V. No cached prompt tokens.
Speed = total tokens / total measured seconds, including JIT.

Standard prompts · prompt speed

Prompt tokens per second

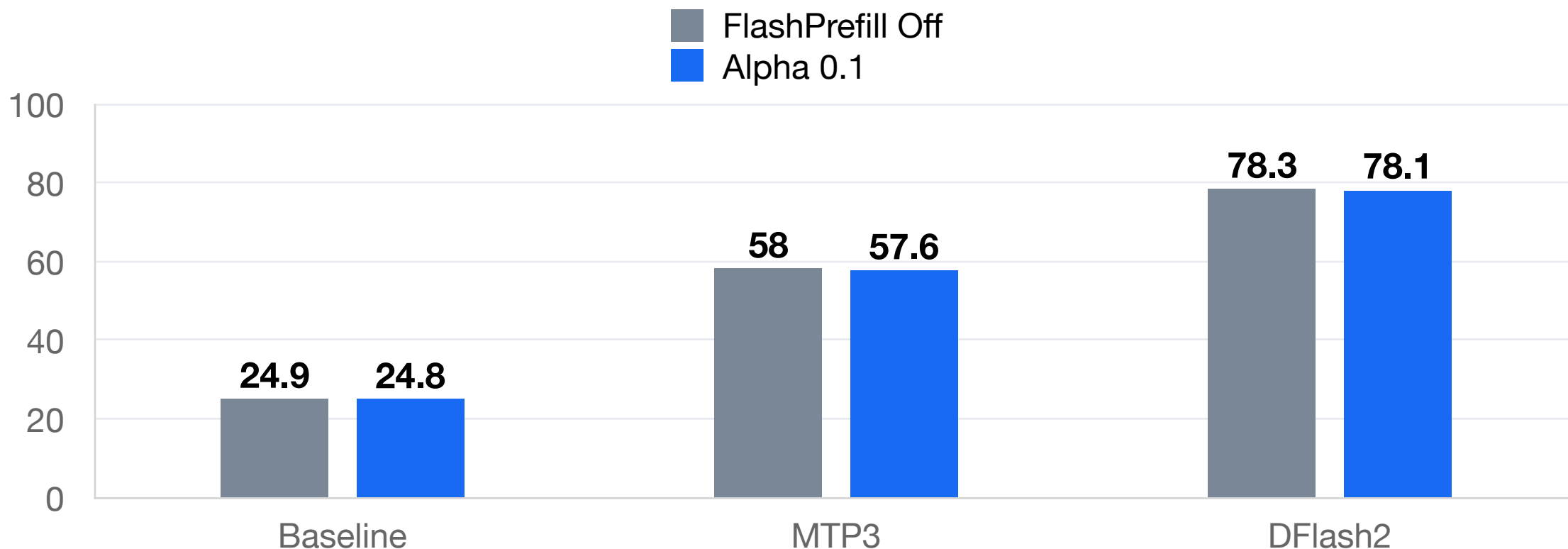


Standard prompts are a separate workload; they are not a zero-context measurement.

164 tasks per setting · Short prompts, with the same task IDs and prompt bytes in each configuration.

Standard prompts · decode speed

Generated tokens per second



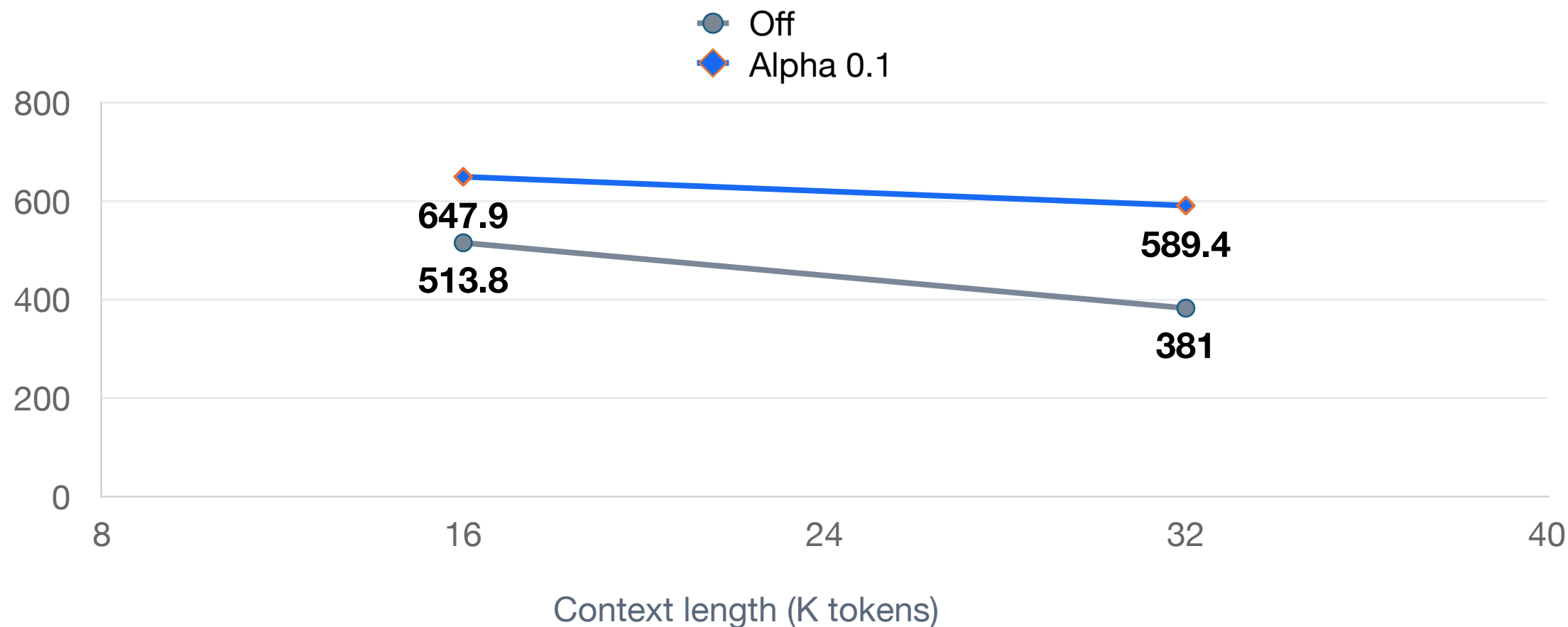
MTP3 and DFlash2 accelerate generation through speculation. Decode attention stays dense.

164 tasks per setting · Short prompts, with the same task IDs and prompt bytes in each configuration.

Baseline · long-context prefill

Prompt tokens per second

+54.7% at 32K

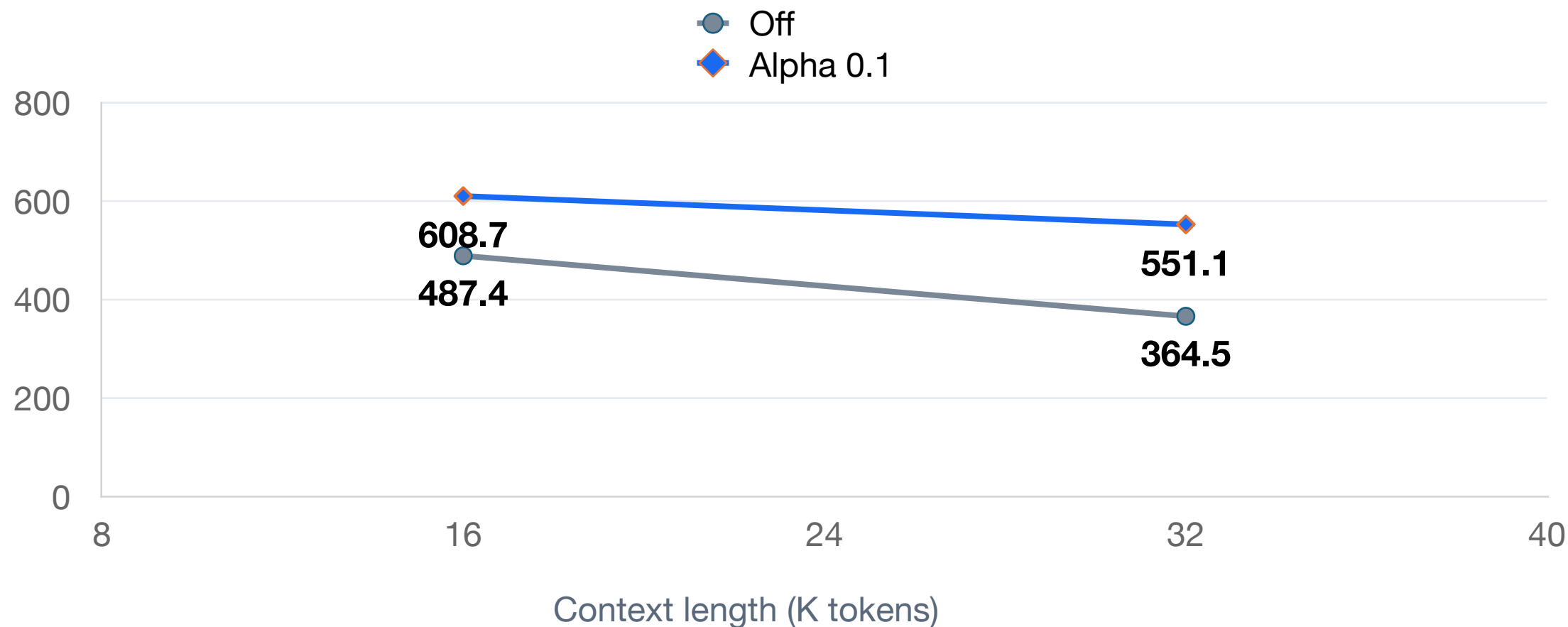


32 tasks per setting · Lines connect two measured points. No intermediate context lengths were tested.

MTP3 · long-context prefill

Prompt tokens per second

+51.2% at 32K

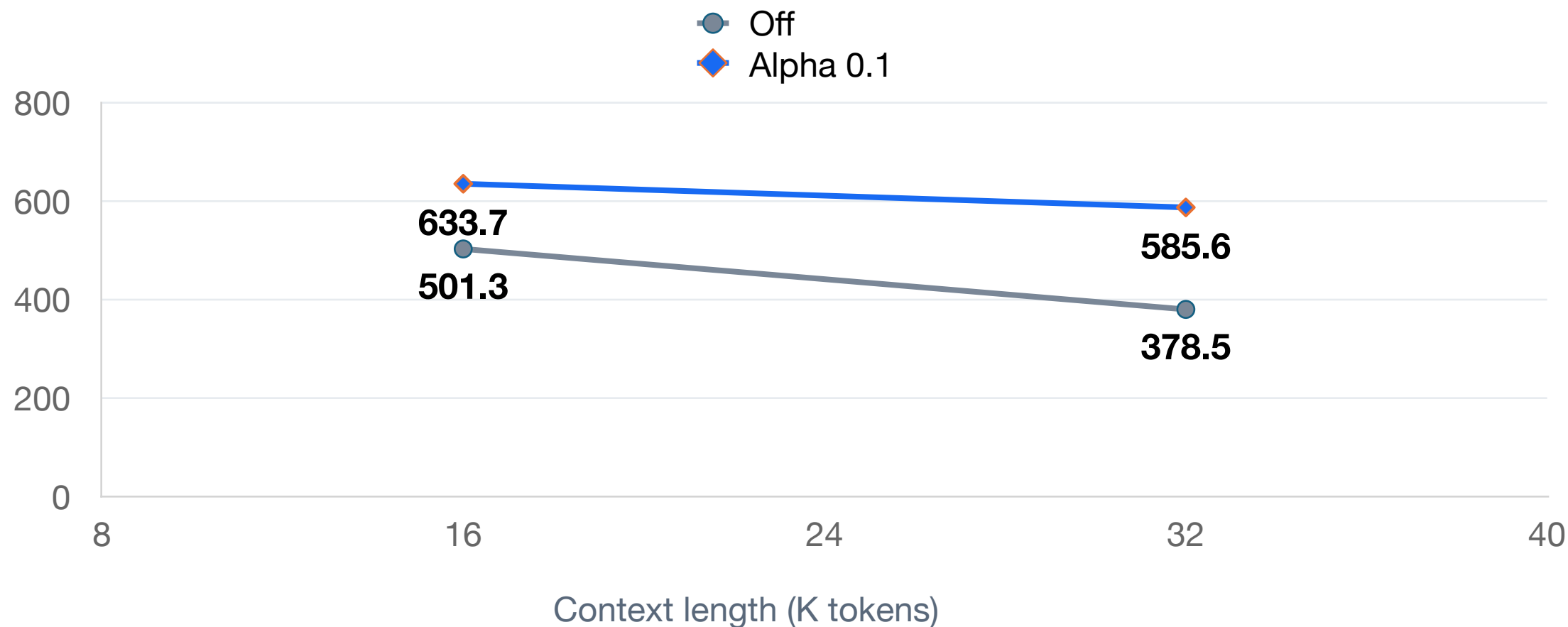


32 tasks per setting · Lines connect two measured points. No intermediate context lengths were tested.

DFlash2 · long-context prefill

Prompt tokens per second

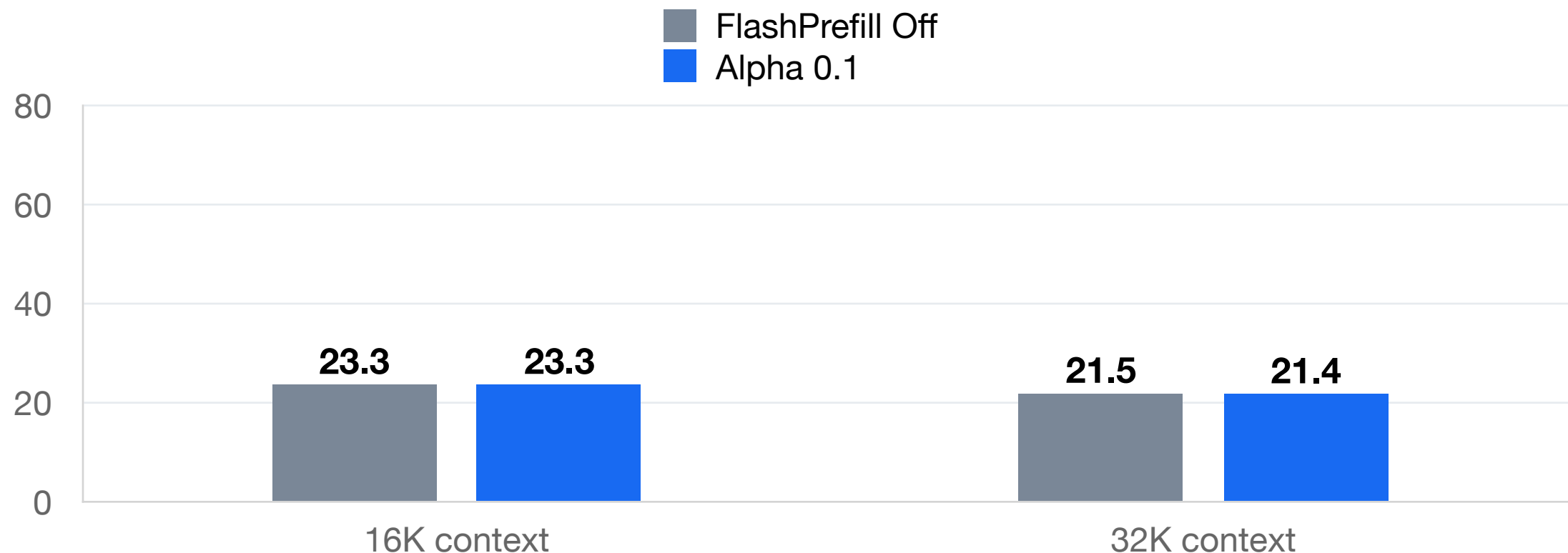
+54.7% at 32K



32 tasks per setting · Lines connect two measured points. No intermediate context lengths were tested.

Baseline · long-context decode

Generated tokens per second

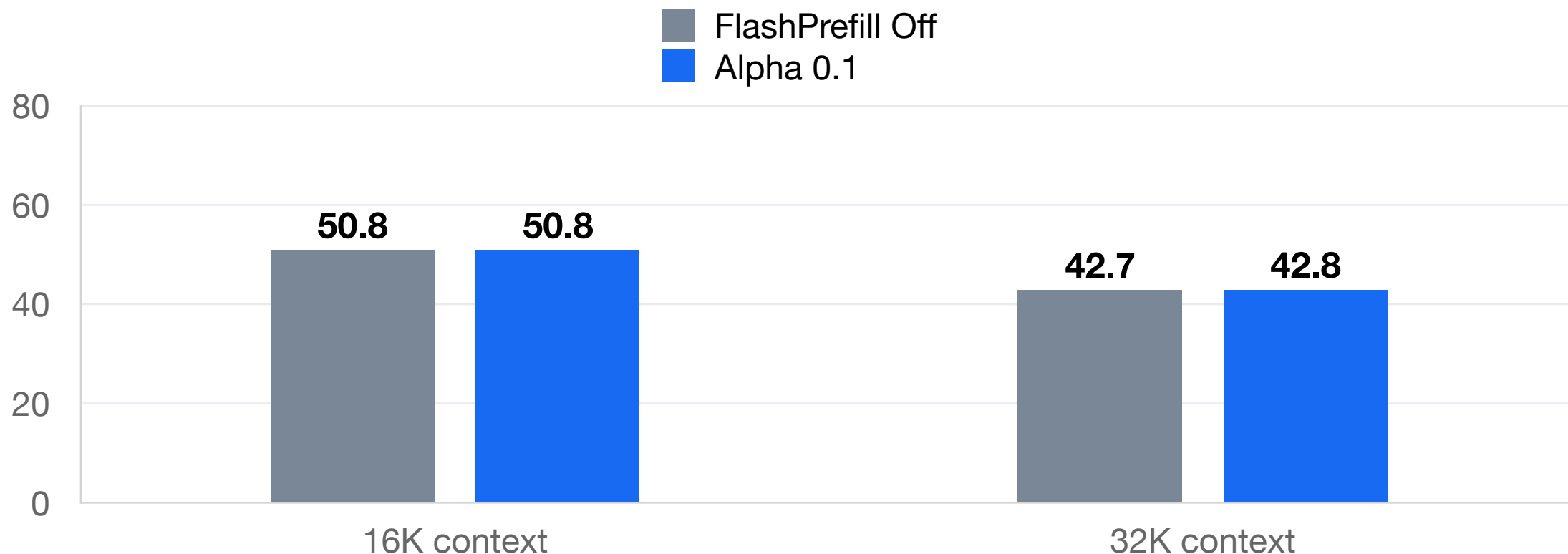


The two settings have very similar decode throughput; FlashPrefill changes prefill attention.

32 tasks per setting · Generation speed also depends on output length and speculative acceptance.

MTP3 · long-context decode

Generated tokens per second

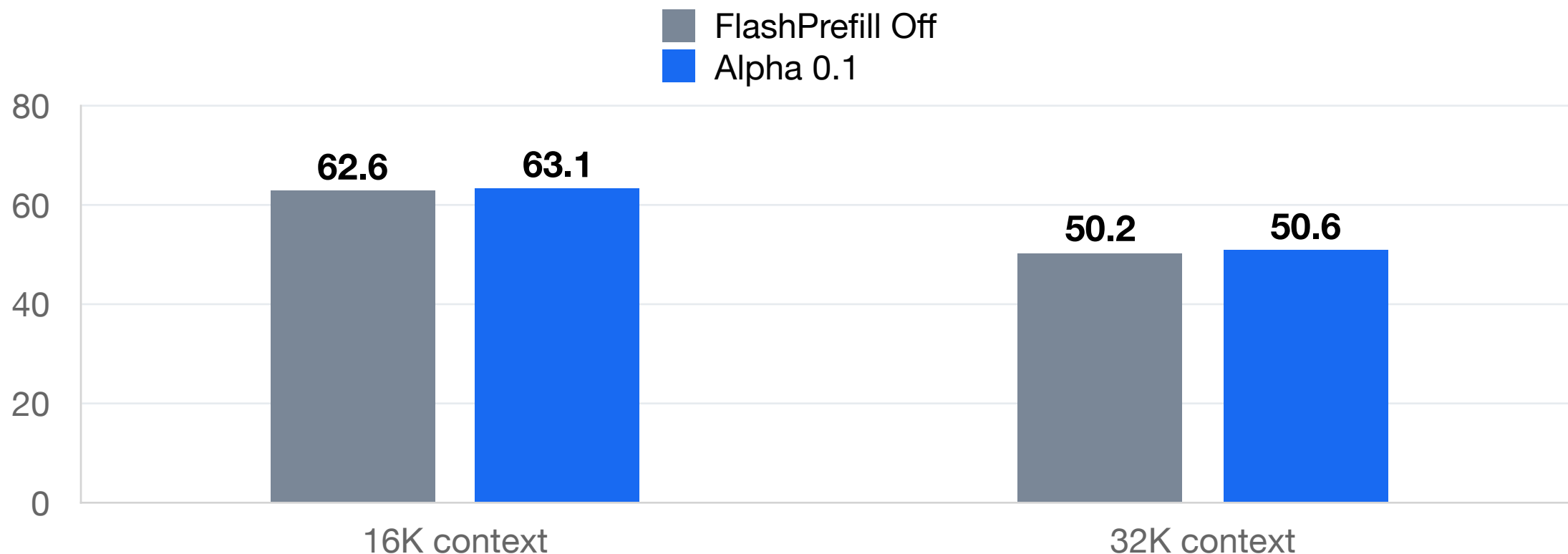


The two settings have very similar decode throughput; FlashPrefill changes prefill attention.

32 tasks per setting · Generation speed also depends on output length and speculative acceptance.

DFlash2 · long-context decode

Generated tokens per second

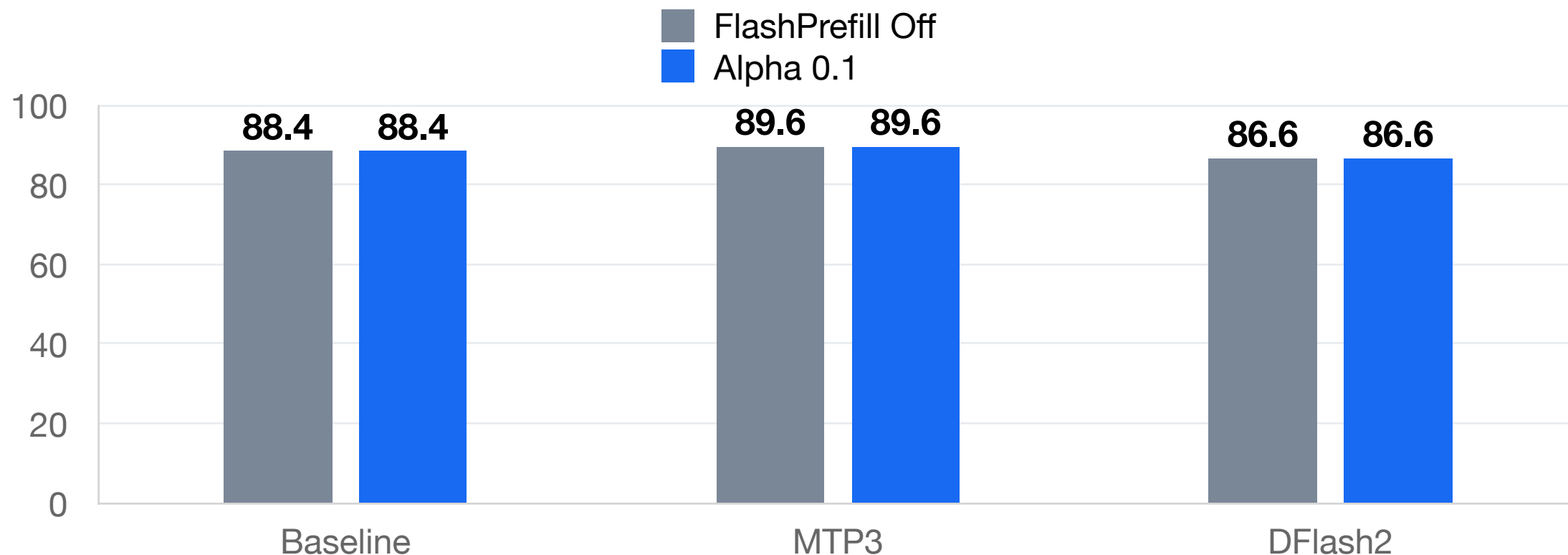


The two settings have very similar decode throughput; FlashPrefill changes prefill attention.

32 tasks per setting · Generation speed also depends on output length and speculative acceptance.

Standard HumanEval+ · correctness

Programs passing every base and extended test (%)



145 / 164 Baseline

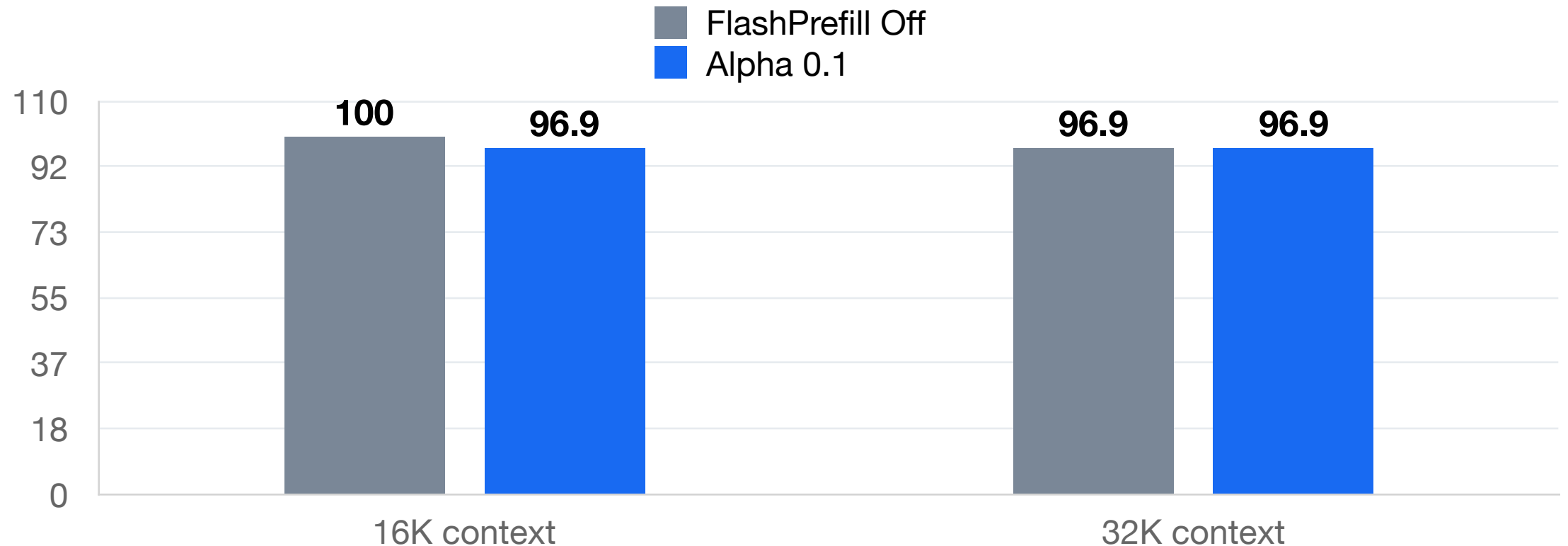
147 / 164 MTP3

142 / 164 DFlash2

164 tasks per setting · Identical pass/fail task IDs for Off and alpha 0.1 within each mode.

Long-context correctness

Programs passing every base and extended test (%)



16K: 32 / 32 Off, 31 / 32 alpha 0.1 • 32K: both 31 / 32

The same outcomes occurred in Baseline, MTP3 and DFlash2. Each result uses the same fixed 32 tasks.

One task changed at 16K

HumanEval/97

Off passed. Alpha 0.1 failed.
The change occurred in all three inference modes.

Standard

0 changed outcomes

16K

1 pass → fail per mode

32K

0 changed outcomes